

Rethinking Entropy Interventions in RLVR: An Entropy Change Perspective

重新思考 RLVR 中的熵干预：从熵变的角度出发

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Abstract

摘要

Reinforcement Learning with Verifiable Rewards (RLVR) serves as a cornerstone technique for enhancing the reasoning capabilities of Large Language Models (LLMs). However, its training is

often plagued by entropy collapse, a rapid decline in policy entropy that limits exploration and undermines training effectiveness. While recent works attempt to mitigate this issue via several heuristic entropy interventions, the underlying mechanisms remain poorly understood. In this work, we conduct comprehensive theoretical and empirical analyses of entropy dynamics in RLVR, offering two main insights: (1) We derive a tight analytical approximation for token-level entropy change at each update step, revealing four governing factors and providing a unified theoretical framework to explain how existing methods influence entropy; (2) We reveal a fundamental limitation of recent approaches: they rely on heuristic adjustments to one or two of these factors, leaving other relevant factors unconsidered, thus inherently limiting their effectiveness. Motivated by these findings, we propose STEER, a principled entropy-modulation method that adaptively reweights tokens based on theoretically-estimated entropy variations. Extensive experiments across six mathematical reasoning and three coding benchmarks demonstrate that STEER effectively mitigates entropy collapse and consistently outperforms state-of-the-art baselines.¹

基于可验证奖励的强化学习（RLVR）是增强大型语言模型（LLMs）推理能力的核心技术。然而，其训练过程常常受到熵坍缩问题的困扰，即策略熵的快速下降会限制探索并削弱训练效果。尽管近期的一些研究尝试通过几种启发式熵干预来缓解这一问题，但其底层机制仍然理解不足。在本工作中，我们对 RLVR 中的熵动态进行了全面的理论和实证分析，提出了两个主要见解：(1) 我们推导出一个针对每次更新步骤中 token 级熵变化的紧致分析近似，揭示了四个主导因素，并提供了一个统一的理论框架，解释现有方法如何影响熵；(2) 我们揭示了近期方法的一个根本性局限：它们依赖于对其中一个或两个因素的启发式调整，而忽略了其他相关因素，从而本质上限制了其有效性。基于这些发现，我们提出了 STEER，这是一种基于原理的熵调节方法，它根据理论上估计的熵变化，自适应地重新加权 token。在六个数学推理和三个编码基准上的广泛实验表明，STEER 有效缓解了熵坍缩，并且在多个基准上持续优于最先进的基线方法。¹

1 Introduction

1 引言

Large Language Models (LLMs) have recently demonstrated remarkable reasoning capabilities in complex tasks such as mathematics and coding (OpenAI, 2025; Google, 2025; Anthropic,

大型语言模型（LLMs）最近在数学和编程等复杂任务中展现出显著的推理能力（OpenAI, 2025; Google, 2025; Anthropic,

2025; Zhang et al., 2025a). A key technique driving these advances is Reinforcement Learning with Verifiable Rewards (RLVR) (Shao et al., 2024; Team et al., 2025). RLVR employs policy-gradient algorithms (e.g., PPO (Schulman et al., 2017) and GRPO (Shao et al., 2024)) guided by automatically verifiable reward signals. This paradigm has been instrumental in unlocking the post-training scaling of reasoning performance (Jaech et al., 2024; Shao et al., 2024; Liu et al., 2025a), serving as a fundamental technique for the latest generation of advanced reasoning models.

2025; Zhang 等, 2025a)。推动这些进展的关键技术是可验证奖励的强化学习（RLVR）(Shao 等, 2024; Team 等, 2025)。RLVR 采用由可自动验证的奖励信号引导的策略梯度算法（例如 PPO (Schulman 等, 2017) 和 GRPO (Shao 等, 2024)）。这一范式在解锁训练后推理性能的扩展方面发挥了重要作用 (Jaech 等, 2024; Shao 等, 2024; Liu 等, 2025a)，成为最新一代高级推理模型的基本技术。

Despite its success, recent studies have identified a major pitfall of RLVR: entropy collapse, a rapid decline in policy entropy during training (Wu and Choi, 2025; Song et al., 2025; Li et al., 2025b; Cui et al., 2025b). This collapse degrades training in two critical ways: (1) It severely restricts exploration, causing the model to generate homogeneous reasoning trajectories (rollouts), thereby failing to discover informative and potentially correct solutions. (2) It disrupts optimization. Particularly for advanced group-based algorithms like GRPO (Shao et al., 2024), the computed advantages become less discriminative, causing training to stagnate. This raises an important research question: How can policy entropy be effectively modulated to preserve exploration?

尽管取得了成功，最近的研究已经指出了 RLVR 的一个主要缺陷：熵坍缩 (entropy collapse)，即在训练过程中策略熵迅速下降 (Wu and Choi, 2025; Song et al., 2025; Li et al., 2025b; Cui et al., 2025b)。这种坍缩以两种关键方式损害了训练效果：(1) 它严重限制了探索，导致模型生成同质化的推理轨迹 (rollouts)，从而无法发现具有信息量且可能正确的解决方案。(2) 它破坏了优化过程。特别是对于像 GRPO (Shao et al., 2024) 这样的高级基于群体的算法，计算出的优势变得不那么具有区分性，导致训练停滞。这引发了一个重要的研究问题：如何有效地调节策略熵以保持探索？

Several recent methods for tackling entropy collapse can be broadly classified into three categories: (1) Clip-Higher, which increases the upper bound of the importance-sampling ratio (Yu et al., 2025; Yang et al., 2025a); (2) Positive-Reweighting, which down-weights positive samples with high generation probability (Zhu et al., 2025; He et al., 2025a); (3) Entropy-Aware Advantage, which assigns larger advantages to tokens with high entropy (Cheng et al., 2026; Tan and Pan, 2025; Wang et al., 2025c,b; Deng et al., 2025). While these strategies empirically mitigate the collapse to some extent, they remain fundamentally heuristic, leaving

几种最近用于解决熵坍缩的方法可以大致分为三类：(1) Clip-Higher，该方法通过提高重要性采样比率的上限来应对熵坍缩 (Yu 等, 2025; Yang 等, 2025a); (2) Positive-Reweighting，该方法通过降低高生成概率的正样本的权重来应对熵坍缩 (Zhu 等, 2025; He 等, 2025a); (3) Entropy-Aware Advantage，该方法通过赋予高熵标记更大的优势来应对熵坍缩 (Cheng 等, 2026; Tan 和 Pan, 2025; Wang 等, 2025c,b; Deng 等, 2025)。尽管这些策略在一定程度上实证地缓解了坍缩问题，但它们本质上仍然是启发式的，留

ing the underlying mechanisms of their impact on entropy largely opaque. Consequently, their effectiveness is bounded, and entropy remains only loosely controlled. These limitations call for a principled understanding of entropy dynamics in RLVR — one that not only demystifies existing heuristics, but also guides the design of more effective entropy-modulation strategies.

理解它们对熵的影响的底层机制在很大程度上是模糊的。因此，它们的效果是有限的，熵的控制也仅是松散的。这些限制要求对 RLVR 中的熵动态有一个原则性的理解——这种理解不仅要揭开现有启发式方法的神秘面纱，还要指导设计更有效的熵调节策略。

Towards this end, this work conducts comprehensive theoretical and empirical analyses of entropy dynamics in RLVR. Moving beyond conventional coarse-grained global expectations, we scrutinize fine-grained token-level entropy variations at each training step. While Cui et al. (2025b) offers a preliminary exploration of entropy, their analyses rely on unrealistic assumptions that lead to highly imprecise estimates (cf. Section 3, Remark 1) and cannot explain the mechanisms underlying existing entropy-intervention methods. In contrast, our analysis yields two main insights: (1) We derive a precise analytical approximation for token-level entropy change, revealing that it is governed by four key factors — clipping strategy, advantage, token probability, and conditional entropy. We further clarify how existing methods influence these factors, thereby explaining their empirical success. (2) We identify key limitations of prior methods: their heuristic modulations are applied only to a small subset of tokens, neglecting others that may also experience severe entropy collapse; and they consider only partial relevant factors, which reduces their effectiveness and can even accelerate entropy collapse in some cases.

为此，本工作对 RLVR 中的熵动态进行了全面的理论和实证分析。超越传统的粗粒度全局期望，我们仔细考察了每个训练步骤中细粒度的 token 级熵变化。虽然 Cui 等人 (2025b) 对熵进行了初步探索，但他们的分析依赖于不现实的假设，导致高度不精确的估计 (参见第 3 节，备注 1)，并且无法解释现有熵干预方法背后的机制。相比之下，我们的分析得出两个主要见解：(1) 我们推导出一个精确的 token 级熵变化分析近似值，揭示了其受四个关键因素的支配——裁剪策略、优势、token 概率和条件熵。我们进一步阐明了现有方法如何影响这些因素，从而解释了它们的实证成功。(2) 我们识别了先前方法的关键局限性：它们的启发式调节仅应用于一小部分 token，忽略了其他可能也经历严重熵崩溃的 token；并且它们只考虑了部分相关因素，这降低了其有效性，甚至在某些情况下会加速熵崩溃。

Motivated by these insights, we introduce STEER (Stabilizing Token-level Entropy-change via Reweighting), a simple yet principled entropy-modulation method grounded in our theoretical analysis. STEER directly translates the theoretically-estimated entropy variations into adaptive token-level weights. By down-weighting tokens with excessively large entropy change, it enables

fine-grained modulation of step-wise entropy dynamics, and effectively steers the policy toward sustained exploration. Extensive experiments across six mathematical reasoning and three coding benchmarks demonstrate that STEER consistently outperforms state-of-the-art baselines by a substantial margin. Importantly, STEER also generalizes well across model scales (1.5B/7B/14B), model families (Qwen/Llama/Mistral), and RL algorithms (GRPO/RLOO/OPO), consistently maintaining stable entropy dynamics and achieving stronger per-

受这些见解的启发，我们引入 STEER（通过重加权实现稳定化词级熵变，STEER），这是一种基于我们理论分析的简单而有原则的熵调节方法。STEER 直接将理论上估计的熵变化转化为自适应的词级权重。通过降低具有过大熵变化的词的权重，它能够实现对逐步熵动态的精细调节，并有效引导策略向持续探索的方向发展。在六个数学推理和三个编码基准测试中进行的大量实验表明，STEER 在很大程度上优于最先进的基线方法。重要的是，STEER 在不同模型规模（1.5B/7B/14B）、模型家族（Qwen/Llama/Mistral）和强化学习算法（GRPO/RLOO/OPO）之间也表现出良好的泛化能力，始终维持稳定的熵动态，并实现更强的每步性能。

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In summary, our main contributions are:

总之，我们的主要贡献是：

- We derive a tight analytical approximation for token-level entropy dynamics, revealing four governing factors and thereby demystifying the mechanisms of existing heuristic entropy-intervention strategies.
- We propose STEER, a theoretically grounded entropy-modulation method that adaptively down-weights tokens susceptible to drastic entropy decay.
- We comprehensively evaluate STEER across six mathematical reasoning and three coding benchmarks, demonstrating its superiority over strong baselines and generalization across varying models and RL algorithms.
- 我们推导出一种紧致的分析近似方法，用于描述令牌级别的熵动态，揭示了四个主导因素，从而揭开现有启发式熵干预策略的机制。
- 我们提出了 STEER，这是一种基于理论的熵调节方法，能够自适应地降低容易导致熵剧烈下降的令牌权重。
- 我们在六个数学推理和三个编码基准上全面评估了 STEER，展示了其在强基线上的优越性，并证明了其在不同模型和强化学习算法上的泛化能力。

2 Preliminaries

2 预备知识

2.1 RLVR Algorithms

2.1 RLVR 算法

Traditional algorithms like PPO (Schulman et al., 2017) drive policy optimization by maximizing an expected advantage, heavily relying on a separate value network for baseline estimation. In contrast, GRPO (Shao et al., 2024) mitigates memory overhead by eliminating the value model, instead computing relative advantages within a sampled group.

传统算法如 PPO (Schulman 等, 2017) 通过最大化预期优势来驱动策略优化, 严重依赖于单独的价值网络来进行基线估计。相比之下, GRPO (Shao 等, 2024) 通过消除价值模型来减轻内存开销, 而是在采样组内计算相对优势。

Formally, let q be a query sampled from a dataset D , and π_θ be the policy model. GRPO samples a group of responses (rollouts) $\{o_i\}_{i=1}^G$ from a reference policy π_{old} . For the t -th token $o_{i,t}$ within response o_i , the advantage is computed as:

正式地, 设 q 是从数据集 D 中采样的查询, 且 π_θ 是策略模型。GRPO 从参考策略 π_{old} 中采样一组响应 (rollouts) $\{o_i\}_{i=1}^G$ 。对于响应 o_i 中的第 t 个 token $o_{i,t}$, 优势值的计算如下:

$$A_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)}, \quad (1)$$

where $R_i \in \{1, -1\}$ indicates the correctness of o_i . This advantage $A_{i,t}$ typically remains constant across all tokens within the response. Following the token-level formulation (Yu et al., 2025), GRPO maximizes the following objective:

其中 $R_i \in \{1, -1\}$ 表示 o_i 的正确性。这一优势 $A_{i,t}$ 通常在响应中的所有 token 之间保持不变。遵循 token 级别的公式 (Yu 等, 2025), GRPO 最大化以下目标:

$$\mathcal{J}(\theta) = \mathbb{E}_{\substack{q \sim \mathcal{D}, \\ \{o_i\}_{i=1}^G \sim \pi_{old}(\cdot|q)}} \left[\frac{1}{L} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min(r_{i,t} A_{i,t}, \text{clip}(r_{i,t}, 1 - \varepsilon, 1 + \varepsilon) A_{i,t}) \right]. \quad (2)$$

where $r_{i,t} = \frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{old}(o_{i,t}|q, o_{i,<t})}$ denotes the importance sampling ratio and $L = \sum_{i=1}^G |o_i|$ denotes the sum of response lengths within a group. Here we omit the KL divergence term following (Shao et al.,

其中 $r_{i,t} = \frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{old}(o_{i,t}|q, o_{i,<t})}$ 表示重要性采样比率, $L = \sum_{i=1}^G |o_i|$ 表示组内响应长度的总和。在这里, 我们省略了后续 (Shao 等人,

2024), which has been shown to yield better performance and broader exploration (Chu et al., 2026; Hu et al., 2025).

2024), 它已被证明能够提供更好的性能和更广泛的探索 (Chu 等, 2026; Hu 等, 2025)。

2.2 Policy Entropy of LLMs

2.2 大型语言模型的策略熵

Shannon entropy quantifies the uncertainty of a policy model’s action selection given a state (Haarnoja et al., 2018). For LLMs, entropy can be quantified at each token generation step. For each sampled response o_i and its t -th token generation, the token-level entropy under policy π_θ is:

Shannon entropy 量化了在给定状态时，策略模型选择动作的不确定性 (Haarnoja 等, 2018)。对于大语言模型 (LLMs)，可以在每个 token 生成步骤中量化 entropy。对于每个采样的响应 o_i 及其 t -th token 生成，策略 π_θ 下的 token 级 entropy 为：

$$\mathcal{H}(q, o_{i,<t}) = -\mathbb{E}_{a \sim \pi_\theta(\cdot | q, o_{i,<t})} [\log \pi_\theta(a | q, o_{i,<t})]. \quad (3)$$

The (global) policy entropy measures the model’s generation uncertainty over a query dataset, which can be estimated by averaging token-level entropy over sampled responses:

(全局) 策略熵衡量了模型在查询数据集上生成的不确定性，可以通过对采样响应进行平均的 token 级熵来估计：

$$\mathcal{H}(\pi_\theta) = \mathbb{E}_{\substack{q \sim \mathcal{D}, \\ \{o_i\} \sim \pi_\theta(\cdot | q)}} \frac{1}{L} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \mathcal{H}(q, o_{i,<t}). \quad (4)$$

In practice, we may estimate $\mathcal{H}(\pi_\theta)$ using the training dataset and empirically observe that this estimation generalizes well to unseen data (e.g., test sets, cf. Appendix D.1).

在实践中，我们可以使用训练数据集来估计 $\mathcal{H}(\pi_\theta)$ ，并经实证观察到这种估计在未见数据（例如测试集，参见附录 D.1）上具有良好的泛化能力。

Maintaining adequate policy entropy is essential for balancing exploration and exploitation during RL training (Haarnoja et al., 2018; Ziebart et al., 2008). However, RLVR training often suffers from entropy collapse, in which policy entropy drops rapidly (Song et al., 2025; Li et al., 2025b; Cui et al., 2025b). This notorious phenomenon is detrimental in two respects: (1) It severely weakens exploration, limiting the model’s ability to discover informative and potentially correct solutions. (2) It also undermines training effectiveness. Typically, for group-based methods such as GRPO (Shao et al., 2024), increasingly homogeneous rollouts yield relative advantage scores $A_{i,t}$ that are less discriminative, causing training to stagnate.

保持适当的策略熵对于在强化学习 (RL) 训练过程中平衡探索与利用至关重要 (Haarnoja 等, 2018; Ziebart 等, 2008)。然而，RLVR 训练常常会遇到熵坍塌问题，即策略熵迅速下降 (Song 等, 2025; Li 等, 2025b; Cui 等, 2025b)。这种臭名昭著的现象在两个方面都是有害的：(1) 它严重削弱了探索能力，限制了模型发现具有信息量且可能正确的解决方案的能力。(2) 它也损害了训练效果。通常情况下，对于基于群体的方法（如 GRPO (Shao 等, 2024)），越来越同质化的 rollout 会产生相对优势得分 $A_{i,t}$ ，这些得分的区分度较低，导致训练停滞。

3 Analyses on Entropy Dynamics

3 关于熵动态的分析

In this section, we present a comprehensive theoretical and empirical analysis of entropy change. Our investigation is conducted at a fine-grained token level, which enables precise monitoring of

entropy dynamics. Without loss of generality, this section primarily focuses on the representative GRPO algorithm for clarity of exposition. Nevertheless, these theoretical results generalize seamlessly to other

在本节中，我们对熵变进行了全面的理论和实证分析。我们的研究是在细粒度的标记级别上进行的，这使得能够精确监控熵的动态变化。为了阐述的清晰性，不失一般性，本节主要聚焦于具有代表性的 GRPO 算法。然而，这些理论结果可以无缝地推广到其他算法。

advanced RLVR algorithms (e.g., PPO (Schulman et al., 2017), RLOO (Ahmadian et al., 2024) and OPO (Hao et al., 2025); cf. Appendix C.2).

高级的 RLVR 算法（例如 PPO (Schulman 等人, 2017 年)、RLOO (Ahmadian 等人, 2024 年) 和 OPO (Hao 等人, 2025 年)；参见附录 C.2)。

To quantitatively characterize the per-token entropy change after one update, we rewrite the GRPO policy gradient (Eq. (2)) as

为了定量表征一次更新后每个 token 的熵变，我们将 GRPO 策略梯度（公式 (2)）重写为

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\substack{q \sim \mathcal{D}, \\ \{o_i\}_{i=1}^G \sim \pi_{\text{old}}(\cdot|q)}} \left[\frac{1}{L} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \mathbb{I}_{\text{clip}} r_{i,t} A_{i,t} \nabla_{\theta} \log \pi_{\theta}(o_{i,t} | q, o_{i,<t}) \right]. \quad (5)$$

where the clipping indicator I_{clip} is derived from the ratio clipping operation and is defined as:

其中，剪裁指示器 I_{clip} 是由比率剪裁操作得出的，并定义为：

$$\mathbb{I}_{\text{clip}} = \begin{cases} 0, & A_{i,t} > 0 \text{ and } r_{i,t} > 1 + \varepsilon, \\ 0, & A_{i,t} < 0 \text{ and } r_{i,t} < 1 - \varepsilon, \\ 1, & \text{otherwise.} \end{cases} \quad (6)$$

For notational convenience, we denote the state (i.e., context) by $s \triangleq (q, o_{i,<t})$ and abbreviate $\pi_{\theta}(a | s)$ as π_{θ} and $A(s, a)$ as A . Then we derive the following theorem on token-level entropy change.

为了便于表示，我们将状态（即上下文）表示为 $s \triangleq (q, o_{i,<t})$ ，将 $\pi_{\theta}(a | s)$ 简写为 π_{θ} ，将 $A(s, a)$ 简写为 A 。然后我们推导出以下关于 token 级熵变化的定理。

Theorem 1. (Token-level Entropy Change). For a logit-independent policy model π_{θ} , trained using GRPO with a learning rate η , the change of the token-level entropy on state s between two consecutive steps can be approximated as:

定理 1. (标记级熵变化)。 对于一个与 logit 无关的策略模型 π_{θ} ，使用 GRPO 进行训练，学习率设为 η ，则在两个连续步骤之间，状态 s 的标记级熵的变化可以近似为：

$$\Omega(s) \triangleq -\frac{\eta}{L} \mathbb{E}_{\pi_{\theta}(\cdot|s)} \left[\frac{\mathbb{I}_{\text{clip}} A}{\pi_{\text{old}}} \pi_{\theta} (1 - \pi_{\theta}) (\log \pi_{\theta} + \mathcal{H}(s)) \right], \quad (7)$$

and the approximation error is bounded by $O(\eta^2)$.

并且近似误差被限制在 $O(\eta^2)$ 以内。

The proof is presented in Appendix G. The theorem is not specific to GRPO, and also holds for other algorithms (cf. Appendix C.2). We highlight three important remarks:

证明见附录 G。该定理并不特定于 GRPO，也适用于其他算法（参见附录 C.2）。我们强调三个重要的注意事项：

Remark 1 (Accurate Estimator): Notably, in practice, since the learning rate η is typically small ($< 10^{-4}$), Theorem 1 delivers an accurate closed-form estimator of token-level entropy dynamics. In contrast, while prior work (Cui et al., 2025b) also explores entropy change, it assumes a uniform entropy distribution across different queries within the same batch. This assumption is rarely satisfied in practice, which leads to an inaccurate approximation of the ground-truth entropy change.

备注 1 (精确估计器): 值得注意的是，在实践中，由于学习率 η 通常较小 ($< 10^{-4}$)，定理 1 提供了一个精确的闭式解估计器，用于估计 token 级别的熵动态变化。相比之下，尽管先前的工作 (Cui 等, 2025b) 也探讨了熵的变化，但它假设同一批次内不同查询具有相同的熵分布。这一假设在实践中很少成立，导致对真实熵变化的近似不准确。

To quantify, we compare our entropy change estimation with their estimation (denoted as Cov)

为了量化，我们将我们的熵变估计与他们的估计进行比较（称为 Cov）

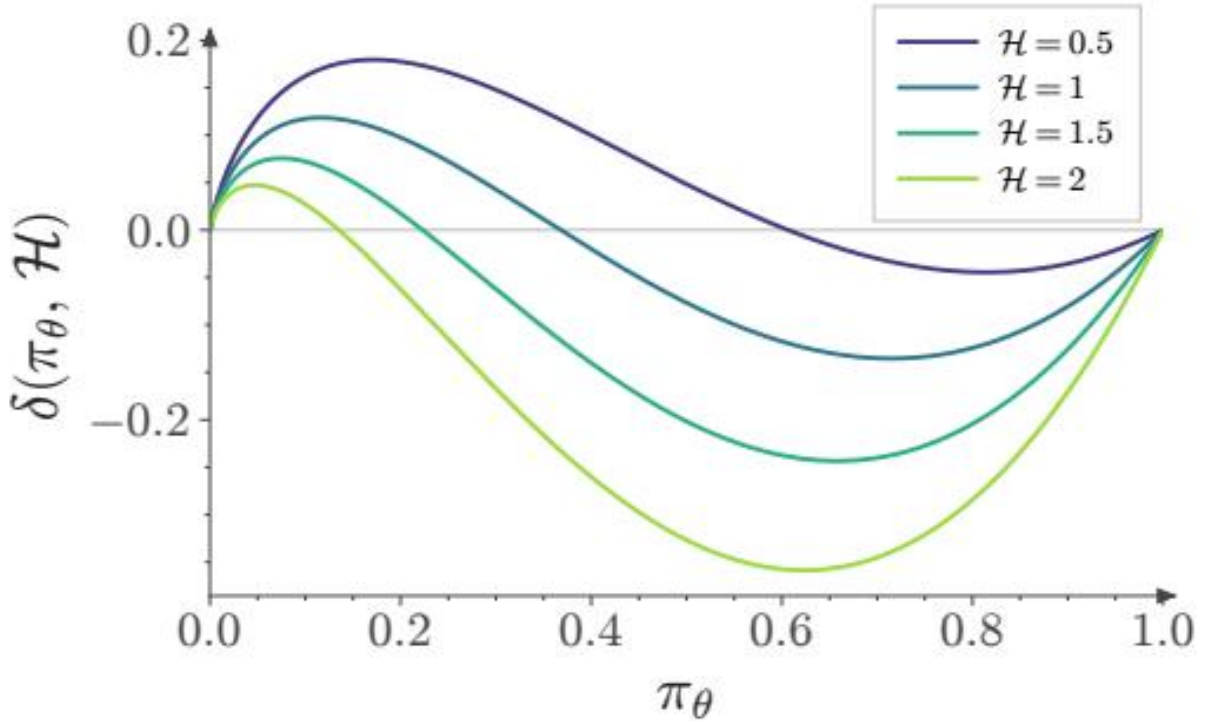


Figure 1: images/de1e8179a81a9d3574bd5d07f2b6dce3274531339206a7edcd88438a0ca1f00b.jpg

Figure 1: δ as a function of π_θ and $\mathcal{H}(s)$.

图 1: δ 作为 π_θ 和 $\mathcal{H}(s)$ 的函数。

Model	Method	MSE	PCC	SRCC
Math-1.5B	Cov	5.37	-6e-5	+0.04
Ours	5e-4	+0.42	+0.65	
Qwen-7B	Cov	0.53	+0.05	+0.08
Ours	8e-4	+0.39	+0.72	
Math-7B	Cov	0.29	+0.03	+0.06
Ours	4e-4	+0.42	+0.61	

模型	方法	MSE	PCC	SRCC
数学-1.5B	Cov	5.37	-6e-5	+0.04
我们的	5e-4	+0.42	+0.65	
Qwen-7B	Cov	0.53	+0.05	+0.08
我们的	8e-4	+0.39	+0.72	
数学-7B	Cov	0.29	+0.03	+0.06
我们的	4e-4	+0.42	+0.61	

Table 1: Accuracy of entropy change estimation: the baseline Cov vs. our estimator.

表 1: 熵变估计的准确性：基线 Cov 与我们的估计器。

during a standard GRPO training process. Table 1 reports the Mean Squared Error (MSE), Pearson Correlation Coefficient (PCC), and Spearman’s Rank Correlation Coefficient (SRCC) between ground-truth entropy change and estimation. Across all three metrics, our method delivers orders-of-magnitude lower MSE and substantially higher PCC and SRCC than Cov. Remarkably, the MSE of our estimator is on the order of $1e-4$, demonstrating its exceptional precision. More results are shown in Appendix D.2.

在标准的 GRPO 训练过程中。表 1 报告了地面真实熵变化与估计值之间的均方误差 (MSE)、皮尔逊相关系数 (PCC) 和斯皮尔曼等级相关系数 (SRCC)。在所有三个指标上，我们的方法比 Cov 的 MSE 低几个数量级，并且 PCC 和 SRCC 显著更高。令人惊讶的是，我们的估计器的 MSE 大约为 $1e-4$ ，这表明其精度非常高。更多结果请参见附录 D.2。

Remark 2 (Four Governing Factors): Theorem 1 demonstrates that token-level entropy change is jointly determined by multiple factors: $\square$ the clip indicator I_{clip} , which prevents the entropy change of the tokens with overly large or small importance sampling ratio; $\square$ the advantage A and the old-policy term π_{old} , which act as weighting factors for entropy change; $\square$ the token generation probability π_θ and $\square$ the token entropy $\mathcal{H}(s)$ of current state. The contribution of π_θ and $\mathcal{H}(s)$ to entropy change can be expressed with the following function:

备注 2 (四个主导因素)：定理 1 表明，token 级别的熵变化是由多个因素共同决定的： $\square$ clip 指示符 I_{clip} ，它防止了具有过大或过小重要性采样比的 token 的熵变化； $\square$ 优势 A 和旧策略项 π_{old} ，它们作为熵变化的权重因子； $\square$ token 生成概率 π_θ 和 $\square$ 当前状态的 token 熵 $\mathcal{H}(s)$ 。 π_θ 和 $\mathcal{H}(s)$ 对熵变化的贡献可以表示为以下函数：

$$\delta(\pi_\theta, \mathcal{H}) \triangleq -\pi_\theta(1 - \pi_\theta)[\log(\pi_\theta) + \mathcal{H}(s)]. \quad (8)$$

The impact of π_θ and $\mathcal{H}(s)$ on the function $\delta(\pi_\theta, \mathcal{H}(s))$ is illustrated in Figure 1.

π_θ 和 $\mathcal{H}(s)$ 对功能 $\delta(\pi_\theta, \mathcal{H}(s))$ 的影响如图 1 所示。

Remark 3 (Entropy Change Direction): We further investigate the direction of entropy change (increase or decrease), which is controlled by the joint effect of the advantage and token probability. Given that the function $\delta(\pi_\theta, \mathcal{H})$ takes negative values for high-probability tokens and positive values

备注 3 (熵变方向): 我们进一步研究熵变的方向 (增加或减少), 这由优势和 token 概率的共同作用所控制。鉴于函数 $\delta(\pi_\theta, \mathcal{H})$ 对于高概率 token 会取负值, 而对低概率 token 会取正值。

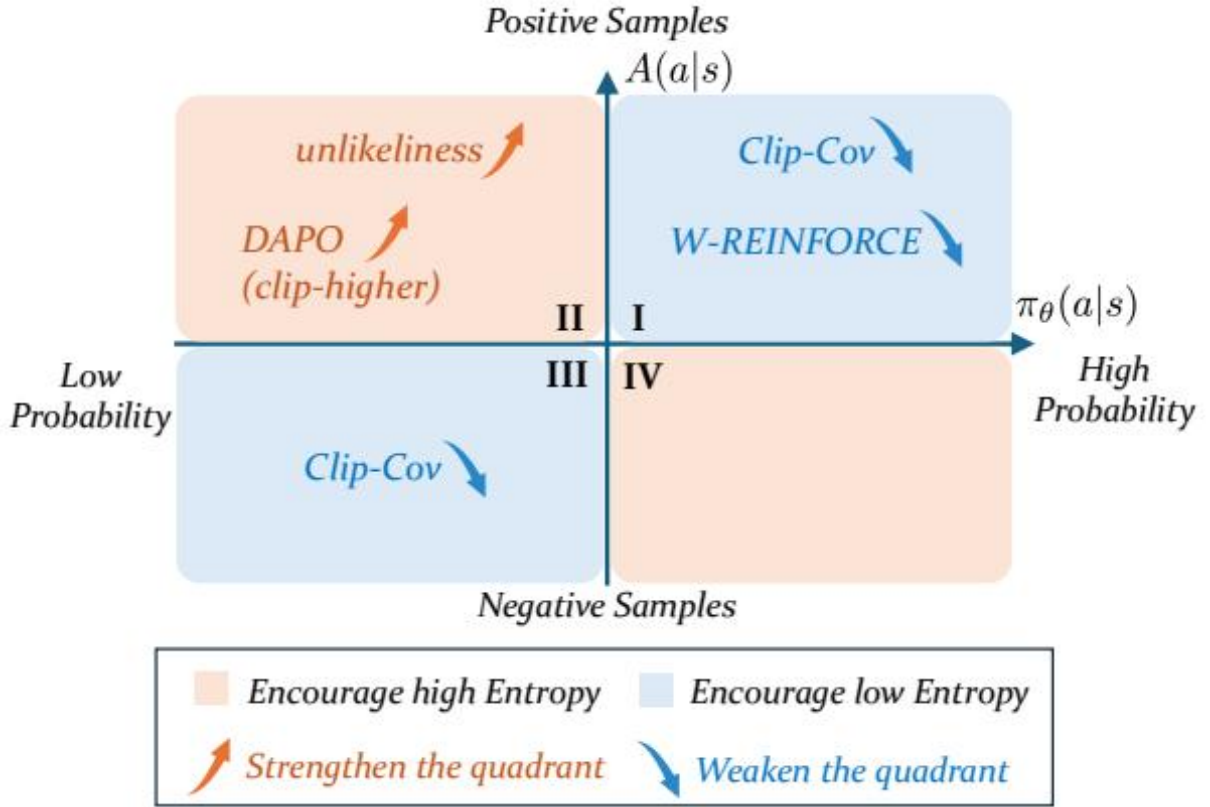


Figure 2: images/d8e3be556eca410551931d239b35f76221fd9b68ebb4cccf8ce8831370833b6b.jpg

Figure 2: Four-quadrant view of token-level entropy change direction with varying advantage and probability. Each token resides in one quadrant, and the macroscopic policy entropy represents the cumulative effect of all quadrants.

图 2: 随着优势和概率变化的 token 级熵变化方向的四象限视图。每个 token 位于一个象限中, 宏观策略熵表示所有象限的累积效应。

for low policy probability, we conceptualize this joint effect in terms of four qualitative quadrants in the (A, π_θ) space shown in Figure 2:

对于低政策概率, 我们以图 2 中所示的 (A, π_θ) 空间中的四个定性象限来概念化这种联合效应:

Quadrant I: Exploitation (entropy decrease). For high-probability tokens in correct outputs ($A > 0$, $\delta < 0$), rewarding a specialized behavior concentrates probability mass, thus decreasing entropy.

第一象限: 利用 (熵减少)。对于正确输出中高概率的标记 ($A > 0$, $\delta < 0$), 奖励一种专门的行为会集中概率质量, 从而减少熵。

Quadrant II: Exploration (entropy increase). For low-probability tokens in correct outputs ($A > 0, \delta > 0$), rewarding a rare-but-correct behavior diversifies the policy, thereby increasing entropy.

第二象限 II：探索（熵值增加）。对于正确输出中的低概率标记（ $A > 0, \delta > 0$ ），奖励罕见但正确的行为可以多样化策略，从而增加熵值。

Quadrant III: Suppression (entropy decrease). For low-probability tokens in incorrect outputs ($A < 0, \delta > 0$), penalizing rare behaviors further concentrates the probability mass, thereby decreasing entropy.

第三象限：抑制（熵减少）。对于错误输出中低概率的标记（ $A < 0, \delta > 0$ ），惩罚罕见行为进一步集中概率质量，从而减少熵。

Quadrant IV: Error-Correction (entropy increase). For high-probability tokens in incorrect outputs ($A < 0, \delta < 0$), penalizing overconfident errors flattens the distribution to encourage seeking alternatives and tends to increase entropy.

第四象限：错误纠正（熵增加）。对于错误输出中高概率的标记 $A < 0, \delta < 0$ ，惩罚过度自信的错误会平滑分布，鼓励寻求替代方案，并倾向于增加熵。

We further perform empirical analyses by intervening on the samples in each quadrant. Our empirical observations are closely aligned with our theoretical findings. Readers may refer to Appendix D.3 for additional details and results.

我们进一步通过对每个象限中的样本进行干预来开展实证分析。我们的实证观察结果与理论发现紧密一致。读者可参考附录 D.3 以获取更多细节和结果。

In the RLVR process, these four quadrant-level dynamics coexist as competing forces that shape the policy. The global policy entropy evolves from the combined effect of these competing updates. Consequently, entropy collapse can be understood as a state where the exploitation-driven, entropy-decreasing updates (Quadrants I and III) consistently overwhelm the exploration-driven, entropy-increasing updates (Quadrants II and IV).

在 RLVR 过程中，这四个象限级别的动态相互共存，作为塑造政策的相互竞争的力量。全局策略熵是从这些竞争更新的综合效应中演变而来的。因此，熵崩溃可以理解作为一种状态，其中以利用为导向、降低熵的更新（象限 I 和 III）持续压倒以探索为导向、增加熵的更新（象限 II 和 IV）。

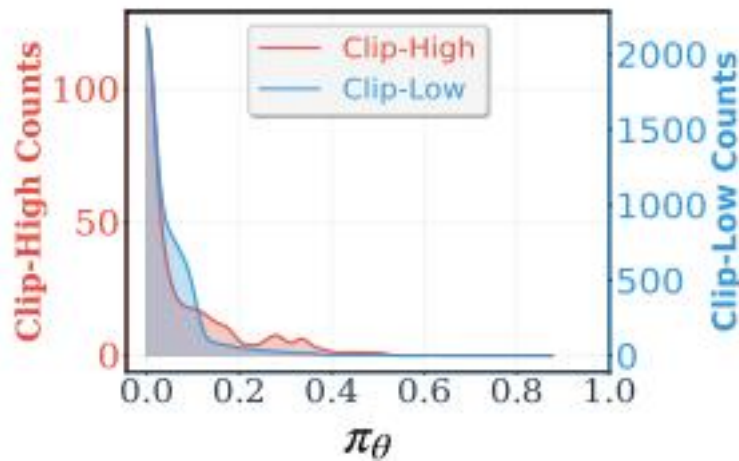


Figure 3: images/0af4ad1ee3a3d6bfa97e6288bf681874b9cf3c805f2c6ad2da54a23bb8adf211.jpg

(a) Clipping statistics

(a) 剪裁统计信息

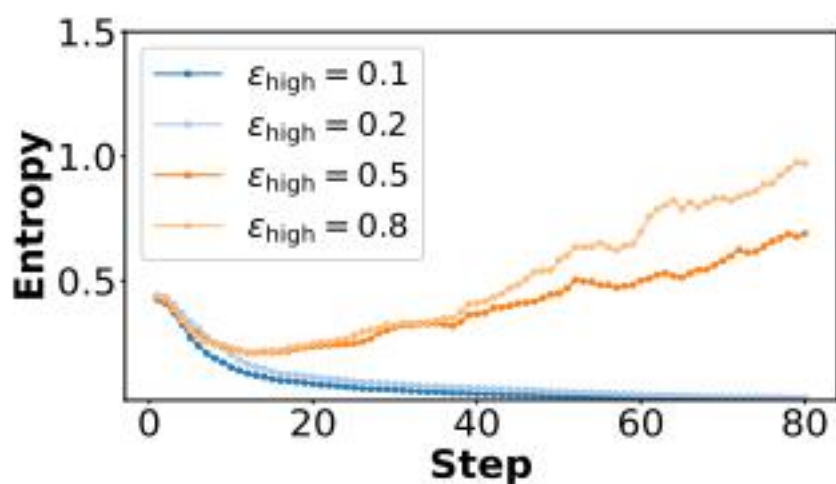


Figure 4: images/a2d123e49fe65d6e7455c7ea2dbcdbe65866e95b787962733c18efc659c403d.jpg

(b) Entropy effect of ϵ_{high}

(b) ϵ_{high} 的熵效应

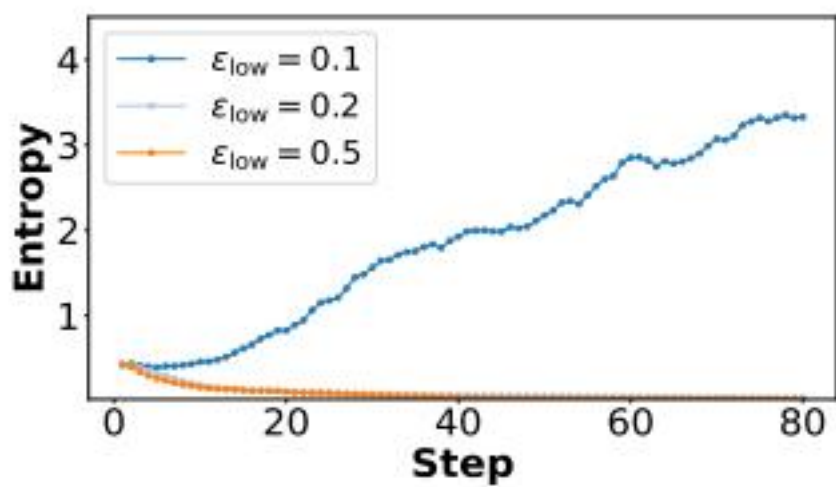


Figure 5: images/5cfb1bb8b059220e6e19836fc1e600e0af77fcd777e0f026a2adc975b7875f27.jpg

(c) Entropy effect of ϵ_{low}

(c) ϵ_{low} 的熵效应

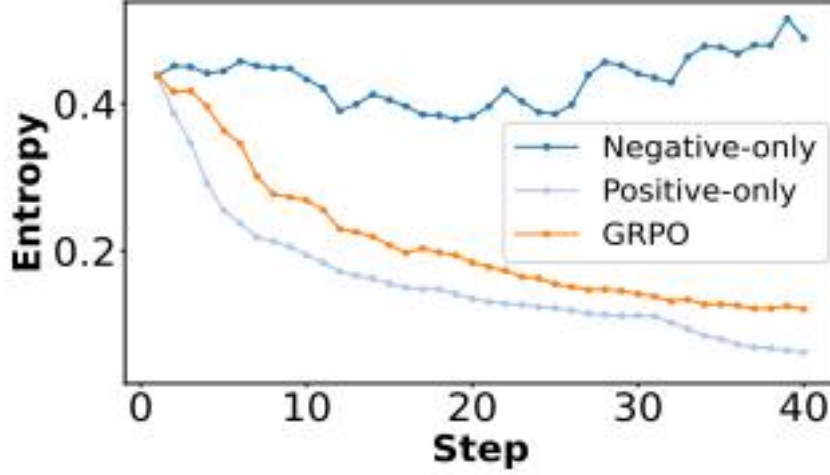


Figure 6: images/4c3858ae4beaec13c0bc7fa2ea944af7ac537e8db49f6a4ab3855e15f9500e12.jpg

(d) Positive/Negative-only

Figure 3: Empirical validation of entropy-change mechanisms of Clip-Higher and Positive-Reweightting.

(d) 只有正样本/只有负样本

图 3: Clip-Higher 和 Positive-Reweightting 的熵变化机制的实证验证。

4 Analyses on Existing Entropy Intervention Methods

4 对现有熵干预方法的分析

Building on the above theoretical findings, we conduct a comprehensive analysis of existing entropy-intervention techniques to reveal their underlying mechanisms and limitations.

基于上述理论发现，我们对现有的熵干预技术进行综合分析，以揭示其内在机制和局限性。

Entropy Effect of Clip-Higher. Recent work (Yu et al., 2025) decouples the lower and higher clipping bounds with ε_{high} and ε_{low} , and shows that increasing ε_{high} can mitigate entropy collapse.

剪辑更高（Clip-Higher）的熵效应。最近的工作（Yu 等，2025）通过 ε_{high} 和 ε_{low} 将较低和较高的剪辑界限解耦，并表明增加 ε_{high} 可以缓解熵坍塌。

Mechanism: The underlying mechanism can be explained as follows: from the importance ratio $r = \frac{\pi_{\theta}}{\pi_{old}}$, the ratio is more likely to attain a large value when π_{old} is relatively small. Consequently, clipping is predominantly triggered for low-probability tokens, a phenomenon confirmed by our empirical observations in Figure 3a, which counts the clipping events in the first 10 steps of GRPO. In this case, the clipping via ε_{high} acts as a filter that removes a considerable number of low-probability positive samples, which typically fall within Quadrant II. Increasing ε_{high} therefore reduces the number of filtered instances, allowing more samples to contribute to entropy increase, thereby mitigating collapse. A similar reasoning applies to the role of ε_{low} : increasing ε_{low} filters fewer instances in Quadrant III which in turn exacerbates entropy collapse.

机制：其基本原理可以解释如下：从重要性比率 $r = \frac{\pi_{\theta}}{\pi_{old}}$ 来看，当 π_{old} 相对较小时，该比率更有可能取得较大的值。因此，裁剪（clipping）主要由低概率的 token 触发，这一现象在图 3a 中得到了我们的实证观察的证实，该图统计了 GRPO 前 10 步中的裁剪事件。在这种情况下，通过 ε_{high} 进行的裁剪起到一

个过滤器的作用，去除了大量低概率的正样本，这些样本通常位于第二象限。因此，增加 ε_{high} 会减少被过滤的实例数量，使得更多的样本能够对熵的增加做出贡献，从而缓解崩溃现象。对于 ε_{low} 的作用，也可以用类似的推理来理解：增加 ε_{low} 会减少第三象限中被过滤的实例数量，从而加剧熵崩溃。

Empirical Evidence: We further validate these observations through empirical experiments. Figure 3b and Figure 3c illustrate the impact of varying ε_{high} and ε_{low} . We observe that increasing ε_{high} can mitigate or even reverse entropy collapse, while increasing ε_{low} intensifies collapse.

实证证据：我们通过实证实验进一步验证了这些观察结果。图 3b 和图 3c 展示了 ε_{high} 和 ε_{low} 变化的影响。我们观察到，增加 ε_{high} 可以缓解甚至逆转熵崩溃，而增加 ε_{low} 则会加剧崩溃。

Limitation: Briefly, the effect of tuning clipping thresholds on entropy dynamics can be interpreted as reweighting tokens in Quadrant II and Quadrant III in Figure 2. However, such heuristic methods remain coarse-grained: DAPO, for example, seeks to affect entropy by controlling the updates of some tokens in Quadrant II, without explicitly controlling tokens in other quadrants.

限制：简而言之，调整截断阈值对熵动态的影响可以解释为对图 2 中第二象限和第三象限中的标记进行重新加权。然而，这样的启发式方法仍然较为粗略：例如，DAPO 试图通过控制第二象限中某些标记的更新来影响熵，而并未明确地控制其他象限中的标记。

Entropy Effect of Re-weighting Positives. Recent studies have shown that re-weighting positive samples can mitigate entropy collapse. For example, unlikelihood (He et al., 2025a) up-weighting tokens in Quadrant II and down-weighting tokens in Quadrant I, while W-REINFORCE (Zhu et al., 2025) down-weighting all positives in training.

重新加权正样本的熵效应。最近的研究表明，重新加权正样本可以缓解熵坍塌。例如，unlikelihood (He 等, 2025a) 对第二象限中的标记进行加权，而对第一象限中的标记进行减权，而 W-REINFORCE (Zhu 等, 2025) 则在训练过程中对所有正样本进行减权。

Mechanism: The effect of unlikelihood is clear from Figure 2: it strengthens the entropy-increasing quadrant and weakens entropy-decreasing quadrant, thereby increasing policy entropy. As for W-REINFORCE, the key insight is that token-level updates are dominated by high-probability tokens since they are more likely to be sampled. Therefore, down-weighting all positives primarily weakens the entropy-decreasing contribution from Quadrant I, thereby increasing policy entropy.

机制：从图 2 可以看出，不可靠性的影响是明显的：它加强了熵增的象限，同时削弱了熵减的象限，从而增加策略熵。至于 W-REINFORCE，关键的见解是，由于高概率的 token 更有可能被采样，因此 token 级别的更新主要由高概率 token 主导。因此，对所有正样本进行降权主要削弱了来自第一象限的熵减贡献，从而增加策略熵。

Empirical Evidence: To validate this mechanism, we conduct positive-only and negative-only experiments in GRPO setting, where the policy is trained using only positive or only negative samples, respectively. The results, shown in Figure 3d, are consistent with the mechanism: training only on positive samples rapidly collapses policy entropy, while training only on negative samples sustains consistently high entropy.

实证证据：为了验证这一机制，我们在 GRPO 设置中进行仅正样本和仅负样本的实验，其中策略分别仅使用正样本或仅使用负样本进行训练。结果如图 3d 所示，与该机制一致：仅使用正样本进行训练会使策略熵迅速降低，而仅使用负样本进行训练则能持续保持较高的熵值。

Limitation: While these positive re-weighting methods improve global entropy, they only reweight a subset of tokens, leaving the remaining tokens still vulnerable to entropy collapse.

限制：虽然这些正面重加权方法可以提高全局熵，但它们仅对一部分标记进行重加权，使得剩余的标记仍然容易受到熵崩溃的影响。

Entropy Effect of Entropy-aware Advantage. Several studies have proposed incorporating

entropy-related terms into the advantage function to mitigate entropy collapse, such as Entro. Adv. (Cheng et al., 2026) and GTPO (Tan and Pan, 2025). Such methods assign larger advantages to tokens with higher entropy. However, our findings indicate that these methods are not universally effective; in fact, they can sometimes exacerbate entropy collapse instead of mitigating it.

熵感知优势的熵效应。已有几项研究提出将与熵相关的术语纳入优势函数中，以缓解熵崩溃，例如 Entro. Adv. (Cheng 等, 2026) 和 GTPO (Tan 和 Pan, 2025)。这些方法会将更大的优势赋予具有更高熵的标记。然而，我们的研究结果表明，这些方法并非在所有情况下都有效；事实上，它们有时甚至会加剧熵崩溃，而非缓解它。

Mechanism: Based on Eq. (8), we plot $\delta(\pi_\theta, \mathcal{H})$ as a function of H in Figure 4a. We can infer that high-entropy tokens tend to induce larger entropy changes. This mechanism is confirmed by empir-

机制：基于式 (8)，我们将 $\delta(\pi_\theta, \mathcal{H})$ 作为 H 的函数绘制在图 4a 中。我们可以推断出高熵词元倾向于引起更大的熵变。这一机制得到了 empir- 的证实。

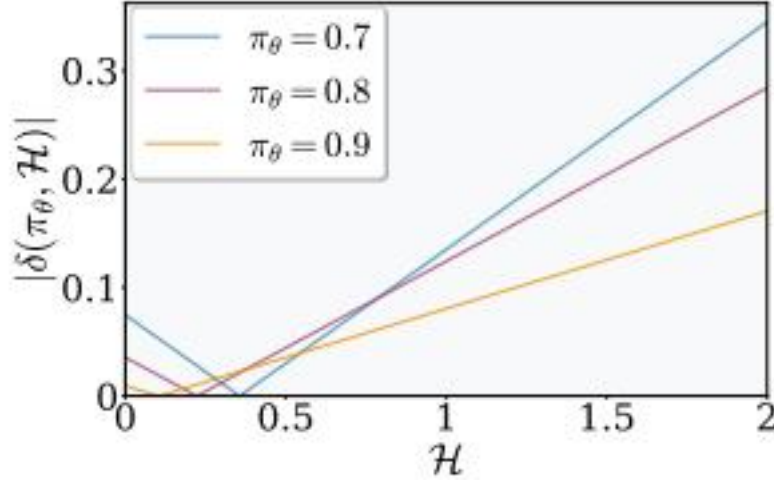


Figure 7: images/f9a5d1635b27183588b8c9948e2a48fe345834fe9a0b966862acabfd1c82a317.jpg

(a) Theoretical correlation

(a) 理论关联

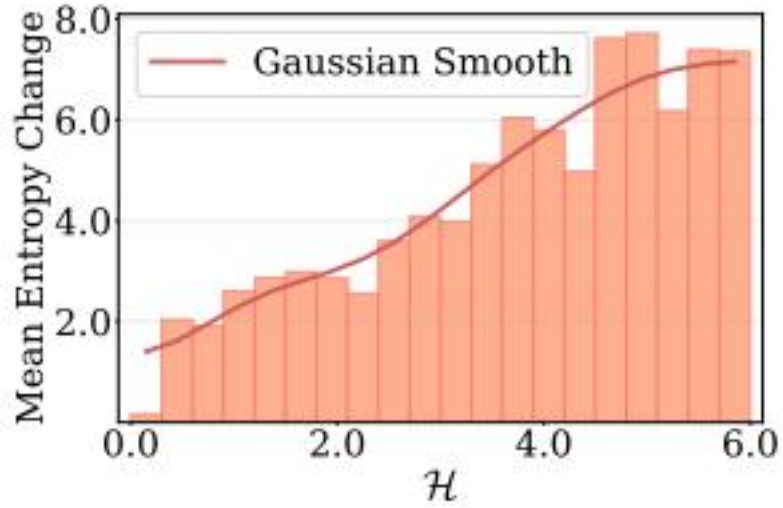


Figure 8: images/d25f6919cb1082037edd2cb66c5d494706fa5a1dc161e0bf48e2e8540eda251d.jpg

(b) Empirical behavior

(b) 实证行为

Figure 4: Entropy change as a function of entropy.

图 4：熵变随熵的变化关系。

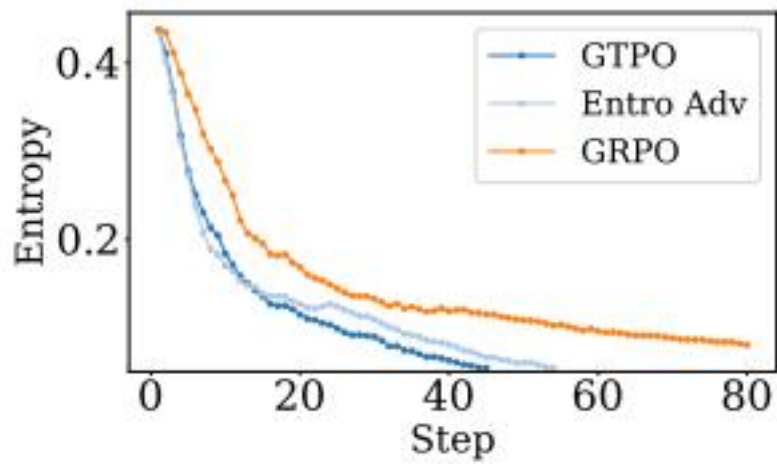


Figure 9: images/2d5543e05521cbf986742ab06694f9f0abc43d49c51feff445943968688b0d94.jpg

(a) On dataset DAPO-17k

(a) 在 DAPO-17k 数据集上

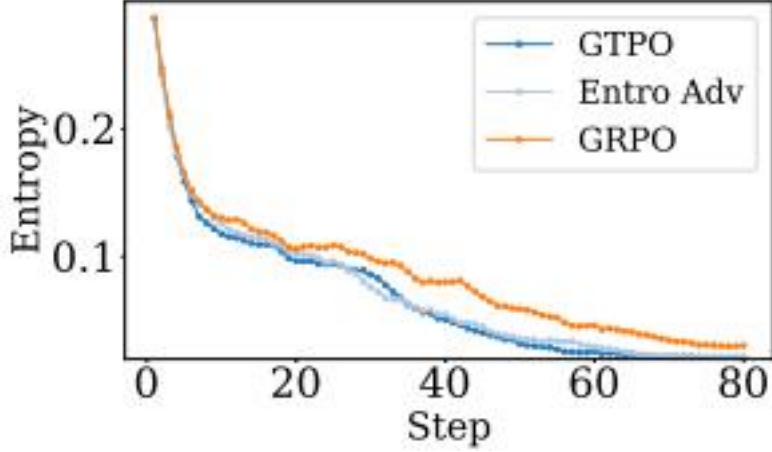


Figure 10: images/9185e61e5e8d50b452329dd373bad3262dc8b85593496b3e13cc308fd4536999.jpg

(b) On dataset Math

Figure 5: Entropy dynamics with entropy-aware advantage on Qwen2.5-Math-7B backbone.

(b) 在 Math 数据集上

图 5: 在 Qwen2.5-Math-7B 主干网络上使用熵感知优势的熵动态变化。

ical behavior during training in Figure 4b, which tracks the mean entropy change over the first 50 training steps: tokens with higher entropy exhibit larger entropy changes. Consequently, assigning greater advantages to high-entropy tokens can amplify their entropy change. When the policy enters an entropy-decreasing phase, this amplification exacerbates the collapse rather than mitigating it.

在图 4b 中，我们观察了训练期间的熵变化行为，该图跟踪了前 50 个训练步骤中平均熵的变化：熵值较高的 token 表现出更大的熵变化。因此，赋予高熵 token 更大的优势可能会放大其熵变化。当策略进入熵减少阶段时，这种放大作用会加剧崩溃现象，而非缓解它。

Empirical Evidence: We validate this mechanism by examining the entropy dynamics of these methods in Figure 5: compared to GRPO, Entro. Adv. and GTPO both exhibit faster entropy collapse when the policy enters an entropy-decreasing phase. When policy entropy starts to decline, these methods can even accelerate entropy collapse.

实证证据：我们通过图 5 中这些方法的熵动态来验证这一机制：与 GRPO 相比，Entro. Adv. 和 GTPO 在策略进入熵减少阶段时表现出更快的熵坍塌。当策略熵开始下降时，这些方法甚至可以加速熵坍塌。

Limitation: This finding highlights a flaw in these methods: rather than reliably encouraging exploration, they may instead aggravate entropy decline.

限制：这一发现指出了这些方法的一个缺陷：它们本应可靠地鼓励探索，但却可能反而加剧熵值的下降。

Limitations of Existing Methods. While the aforementioned entropy-intervention strategies can partly alleviate entropy collapse in practice, they remain largely heuristic and lack principled guidance. These strategies involve qualitative adjustments to only one or two factors in Eq. (7), as shown in Table 2, leaving other relevant factors unconsidered and failing to capture their joint impacts on the entropy dynamics. As a result, there is a substantial gap between their interventions and the actual entropy evolution observed during training. Their effectiveness is compromised.

现有方法的局限性。虽然上述的熵干预策略在实践中可以部分缓解熵崩溃问题，但它们仍然主要属于启发式方法，并缺乏理论指导。这些策略仅对公式 (7) 中的一个或两个因素进行定性调整，如表 2 所示，而忽略了其他相关因素，未能捕捉到它们对熵动态的联合影响。因此，它们的干预措施与训练过程中实际观察

到的熵演变之间存在显著差距。其有效性因此受到削弱。

5 Stabilizing Token-level Entropy-change via Reweighting

5 通过重加权稳定令牌级熵变

The preceding analysis motivates us to develop a fine-grained and theoretically-grounded method, termed STEER. The key idea is to directly leverage

前述分析促使我们开发一种细粒度且有理论依据的方法，称为 STEER。其关键思想是直接利用

Method	$\mathbb{I}_{clip}$	π_θ	A	$\mathcal{H}(s)$
DAPO	□	□	□	□
Unlikelihood	□	□	□	□
W-REINFORCE	□	□	□	□
Entro. Adv.	□	□	□	□
KL Reg.	□	□	□	□
Entropy Reg.	□	□	□	□
Edge-GRPO	□	□	□	□
Forking Tokens	□	□	□	□
Clip-Cov	□	□	□	□
STEER	□	□	□	□

方法	$\mathbb{I}_{clip}$	π_θ	A	$\mathcal{H}(s)$
DAPO	□	□	□	□
不可能性	□	□	□	□
W-REINFORCE	□	□	□	□
进入。副词。	□	□	□	□
KL Reg.	□	□	□	□
熵正则化	□	□	□	□
Edge-GRPO	□	□	□	□
分叉代币	□	□	□	□
剪辑覆盖	□	□	□	□
STEER	□	□	□	□

Table 2: Factors that governing entropy change considered by existing entropy-intervention methods and ours. the theoretically-estimated entropy change $\Omega(s)$, translating it into an adaptive token-level scaling weight $\lambda(s)$, incorporated in the loss function for each token. Accordingly, for tokens susceptible to drastic entropy decay, STEER actively attenuates their gradient contribution, approximately modulating their entropy variations from $\Omega(s)$ to $\lambda(s)\Omega(s)$. This mechanism empowers STEER to execute precise token-level entropy modulation, moving beyond the coarse monitoring of global expectations by prior methods.

表 2: 现有熵干预方法和我们方法所考虑的控制熵变化的因素。理论估计的熵变化 $\Omega(s)$ ，将其转换为自适应的 token 级别缩放权重 $\lambda(s)$ ，并将其纳入每个 token 的损失函数中。因此，对于容易发生剧烈熵衰减

的 token，STEER 会主动减弱其梯度贡献，从而近似地将它们的熵变化从 $\Omega(s)$ 调节到 $\lambda(s)\Omega(s)$ 。这种机制使 STEER 能够执行精确的 token 级别熵调节，超越了先前方法对全局期望的粗略监控。

To achieve this, we introduce $\lambda(s)$ through a simple exponential decay function:

为了实现这一点，我们通过一个简单的指数衰减函数引入 $\lambda(s)$ ：

$$\lambda(s) = \exp\left(-\alpha \frac{|\Omega(s)|}{\max_{s \in \mathcal{B}} |\Omega(s)|}\right), \quad (9)$$

where $\alpha > 0$ is the hyperparameter controlling the decay rate. This formulation satisfies two desired properties: (1) it is monotonically decreasing with respect to the normalized entropy change; (2) the exponential ensures that all weights remain strictly positive. Here we apply the absolute value $|\Omega(s)|$ to measure the magnitude of entropy-change, since large entropy increases may also be sub-optimal. Controlling both increases and decreases within a stable range leads to more stable training dynamics. The normalization by the batch maximum ensures numerical stability.

其中 $\alpha > 0$ 是控制衰减率的超参数。这种形式满足两个期望的性质：(1) 它随着归一化熵变化单调递减；(2) 指数确保所有权重保持严格正数。在这里我们应用绝对值 $|\Omega(s)|$ 来衡量熵变化的大小，因为较大的熵增加也可能不是最优的。在稳定的范围内控制增减可以带来更稳定的训练动态。通过批次最大值进行归一化以确保数值稳定性。

The hyperparameter α governs the slope of the decay curve. Equivalently, α determines the minimum attainable weight, $\lambda_{\min} = \exp(-\alpha)$. Thus, Eq. (9) can be interpreted as an inverse mapping of entropy changes into the range $[\lambda_{\min}, 1]$, with larger entropy changes corresponding to smaller weights. In practice, we find it more convenient to tune $\lambda_{\min}$ instead of α , due to its better interpretability.

超参数 α 控制衰减曲线的斜率。等价地， α 确定可达到的最小权重， $\lambda_{\min} = \exp(-\alpha)$ 。因此，式 (9) 可以解释为将熵的变化逆映射到范围 $[\lambda_{\min}, 1]$ 内，较大的熵变化对应较小的权重。在实践中，我们发现由于其更好的可解释性，调节 $\lambda_{\min}$ 比调节 α 更为方便。

Method	AIME24	AIME25	AMC23	MATH500	Minerva	Olympiad	Avg.
Qwen2.5-Math-7B	13.8	5.3	44.6	39.6	9.9	13.8	21.2
Classical RLVR Methods							
GRPO	28.0	14.3	66.2	78.6	37.3	40.9	44.2
SimpleRL-Zoo	30.8	14.2	65.4	79.2	37.1	40.8	44.6
Eurus-PRIME	20.9	13.0	65.2	79.8	37.4	40.6	42.8
OPO	32.2	13.4	71.5	82.2	38.2	41.0	46.4
Entropy Intervention Methods							
GRPO w/ clip-high	31.7	12.8	66.8	79.0	38.6	39.3	44.7
GRPO w/ Entro. Loss	29.1	14.0	67.6	80.0	38.2	37.9	44.5
GRPO w/ Fork Tokens	31.9	14.3	65.5	79.2	37.1	40.9	44.8
W-REINFORCE	29.2	12.0	64.9	79.2	37.8	40.9	44.0
Entro. Adv.	27.5	13.5	70.2	79.6	36.8	42.8	45.1
Clip-Cov	32.5	12.9	68.4	78.0	40.8	41.3	45.7
KL-Cov	32.8	14.1	64.2	78.8	37.1	39.4	44.4
STEER (Ours)	36.2	16.1	72.1	82.2	41.7	43.0	48.6

方法	AIME24	AIME25	AMC23	MATH500	弥涅尔瓦	奥林匹克竞赛	平均值
Qwen2.5-Math-7B	13.8	5.3	44.6	39.6	9.9	13.8	21.2
经典 RLVR 方法							
GRPO	28.0	14.3	66.2	78.6	37.3	40.9	44.2
SimpleRL-Zoo	30.8	14.2	65.4	79.2	37.1	40.8	44.6
Eurus-PRIME	20.9	13.0	65.2	79.8	37.4	40.6	42.8
OPO	32.2	13.4	71.5	82.2	38.2	41.0	46.4
熵干预方法							
GRPO w/ clip-high	31.7	12.8	66.8	79.0	38.6	39.3	44.7
GRPO 与 Entro. 损失	29.1	14.0	67.6	80.0	38.2	37.9	44.5
GRPO 与分叉代币	31.9	14.3	65.5	79.2	37.1	40.9	44.8
W-REINFORCE	29.2	12.0	64.9	79.2	37.8	40.9	44.0
进入。副词。	27.5	13.5	70.2	79.6	36.8	42.8	45.1
剪切覆盖	32.5	12.9	68.4	78.0	40.8	41.3	45.7
KL-Cov	32.8	14.1	64.2	78.8	37.1	39.4	44.4
STEER (Ours)	36.2	16.1	72.1	82.2	41.7	43.0	48.6

Table 3: Benchmark results of different methods on Qwen2.5-Math-7B. We report avg@32 for AIME24, AIME25, and AMC23 and avg@1 for other datasets. All results are presented as percentages.

表 3: 不同方法在 Qwen2.5-Math-7B 上的基准结果。我们对 AIME24、AIME25 和 AMC23 报告 avg@32, 对其他数据集报告 avg@1。所有结果均以百分比形式呈现。

Notably, STEER also enjoys significant practical merits, incurring minimal computational cost and allowing for easy integration into existing methods. It only involves the additional calculation of $\Omega(s)$, the complexity of which is negligible. Besides, STEER can be seamlessly applied to various LLM architectures and RL algorithms.

值得注意的是, STEER 还具有显著的实用优势, 其计算成本极低, 易于集成到现有方法中。它仅涉及额外计算 $\Omega(s)$, 其复杂度可以忽略不计。此外, STEER 可以无缝地应用于各种 LLM 架构和 RL 算法。

6 Experiments and Results

6 实验与结果

In this section, we evaluate STEER from the following perspectives. First, we compare STEER against strong baselines on six math reasoning benchmarks and three coding benchmarks. Second, we examine STEER’s entropy control ability in RLVR training. Third, we test STEER’s generalization across multiple model scales, families, and RL algorithms. Further results are deferred to Appendix F.

在本节中, 我们从以下几个角度评估 STEER。首先, 我们在六个数学推理基准和三个编程基准上, 将 STEER 与强大的基线模型进行比较。其次, 我们考察 STEER 在 RLVR 训练中的熵控制能力。第三, 我们测试 STEER 在多个模型规模、家族和强化学习算法上的泛化能力。进一步的结果详见附录 F。

6.1 Experimental Setup

6.1 实验设置

Training. We follow the default training recipe of standard GRPO in verl (Sheng et al., 2025). For fair comparisons, we closely follow the experimental setup used in recent work (Yu et al., 2025; Wang et al., 2025a). The only hyperparameter introduced by STEER, λ_{min} , is set to 0.7. To ensure the reliability, all results reported in the main experiments are averaged over two independent training runs.

训练。我们遵循 verl 中标准 GRPO 的默认训练方案 (Sheng 等人, 2025 年)。为了公平比较, 我们紧密遵循近期工作的实验设置 (Yu 等人, 2025 年; Wang 等人, 2025a)。STEER 引入的唯一超参数 λ_{min} 设置为 0.7。为确保可靠性, 主实验中报告的所有结果均基于两次独立训练运行的平均值。

For math reasoning task, we conduct experiments on four different models, including Qwen2.5-Math-1.5B/7B, Qwen2.5-14B (Yang et al., 2024a) and Llama-3.2-3B (Grattafiori et al., 2024). We train on the widely-used DAPO-Math-17k (Yu et al., 2025), which consists of 17K prompts, each paired with a ground-truth integer answer.

对于数学推理任务, 我们在四个不同的模型上进行了实验, 包括 Qwen2.5-Math-1.5B/7B、Qwen2.5-14B (Yang et al., 2024a) 和 Llama-3.2-3B (Grattafiori et al., 2024)。我们使用广泛使用的 DAPO-Math-17k 进行训练, 该数据集包含 17,000 个提示, 每个提示都与一个真实整数答案配对。

For coding tasks, we conduct comparison experiments on four different models, including Qwen2.5-

对于编码任务, 我们在四个不同的模型上进行对比实验, 包括 Qwen2.5-

Coder-3B/7B/14B (Hui et al., 2024) and Mistral-7B (Jiang et al., 2023). For code generation task, we adopt ArcherCodeR² for training. As for code editing task, since there is no large-scale open-source dataset available, we adopt our in-house practical code editing benchmark for training.

Coder-3B/7B/14B (Hui et al., 2024) 和 Mistral-7B (Jiang et al., 2023)。对于代码生成任务, 我们采用 ArcherCodeR² 进行训练。至于代码编辑任务, 由于目前没有大规模的开源数据集可用, 我们采用内部实际的代码编辑基准进行训练。

Evaluation. For math reasoning task, we evaluate our models and baselines on six widely used mathematical reasoning benchmarks: AIME24, AIME25, AMC23, MATH-500, Minerva Math, and OlympiadBench. For code generation task, we adopt the widely used LiveCodeBench v5 (Jain et al., 2024) for the evaluation of code generation. We report avg@4 for code generation task following (Wang et al., 2025b). For code editing, we evaluate the model on both our internal held-out test split and Zeta (Zed Industries, 2025). We report exact-match accuracy for measuring code edit task. More training and evaluation details of our method and baselines are listed in Appendix E.

评估。对于数学推理任务, 我们在六个广泛使用的数学推理基准数据集上评估我们的模型和基线: AIME24、AIME25、AMC23、MATH-500、Minerva Math 和 OlympiadBench。对于代码生成任务, 我们采用广泛使用的 LiveCodeBench v5 (Jain 等人, 2024) 进行代码生成的评估。我们按照 (Wang 等人, 2025b) 的方式报告代码生成任务的 avg@4。对于代码编辑任务, 我们在我们内部的保留测试集和 Zeta (Zed Industries, 2025) 上评估模型。我们通过精确匹配准确率来衡量代码编辑任务的性能。我们的方法和基线的更多训练和评估细节列在附录 E 中。

Baselines. For a thorough comparison, we compare STEER on math reasoning task against 11 baselines, including standard GRPO (Shao et al., 2024), SimpleRL-Zoo (Zeng et al., 2025), Euris-PRIME (Cui et al., 2025a), OPO (Hao et al., 2025), GRPO with clip-high (Yu et al., 2025), GRPO with entropy loss (Schulman et al., 2017), GRPO with Fork Tokens (Wang et al., 2025c), W-REINFORCE (Zhu et al., 2025), Entro. Adv. (Cheng et al., 2026), Clip-Cov and KL-Cov (Cui et al., 2025b). And we compare STEER with GRPO on coding tasks. For all baselines, the default training hyperparame-

ters in RLVR are con-

基线。为了进行全面的比较，我们在数学推理任务中将 STEER 与 11 种基线进行比较，包括标准 GRPO (Shao 等, 2024)、SimpleRL-Zoo (Zeng 等, 2025)、Eurus-PRIME (Cui 等, 2025a)、OPO (Hao 等, 2025)、带 clip-high 的 GRPO (Yu 等, 2025)、带熵损失的 GRPO (Schulman 等, 2017)、带 Fork Tokens 的 GRPO (Wang 等, 2025c)、W-REINFORCE (Zhu 等, 2025)、Entro. Adv. (Cheng 等, 2026)、Clip-Cov 和 KL-Cov (Cui 等, 2025b)。此外，我们在编码任务中将 STEER 与 GRPO 进行比较。对于所有基线，默认的训练超参数在 RLVR 中是 con-

Method	AIME24	AIME25	AMC23	MATH500	Minerva	Olympiad	Avg.
Base	3.9	2.6	25.8	52.6	15.4	23.0	20.6
GRPO	17.2	13.2	66.3	80.6	38.0	42.2	42.9
OPO	17.8	12.6	68.2	78.6	37.7	42.6	42.9
Entro. Adv.	14.6	9.8	65.6	78.8	36.5	40.9	41.0
Clip-Cov	14.1	13.6	59.8	78.2	38.6	43.2	41.2
STEER	19.3	14.3	70.3	81.4	39.4	46.7	45.2

方法	AIME24	AIME25	AMC23	MATH500	弥涅尔瓦	奥林匹克	平均值
基础	3.9	2.6	25.8	52.6	15.4	23.0	20.6
GRPO	17.2	13.2	66.3	80.6	38.0	42.2	42.9
OPO	17.8	12.6	68.2	78.6	37.7	42.6	42.9
进入。副词。	14.6	9.8	65.6	78.8	36.5	40.9	41.0
剪切-覆盖	14.1	13.6	59.8	78.2	38.6	43.2	41.2
STEER	19.3	14.3	70.3	81.4	39.4	46.7	45.2

Table 4: Benchmark results of different methods on Qwen2.5-14B. We report avg@32 for AIME24, AIME25, and AMC23 and avg@1 for other datasets. All results are presented as percentages.

表 4: 不同方法在 Qwen2.5-14B 上的基准结果。我们对 AIME24、AIME25 和 AMC23 报告 avg@32，对其他数据集报告 avg@1。所有结果均以百分比形式呈现。

Dataset	Method	3B	7B	14B
Internal	GRPO	39.7	40.1	42.6
	STEER	41.2	41.9	45.1
Zeta	GRPO	17.4	22.0	22.3
	STEER	19.3	24.0	24.1
LCB-v5	GRPO	24.4	28.5	29.3
	STEER	24.9	29.2	31.8

数据集	方法	3B	7B	14B
内部	GRPO	39.7	40.1	42.6
	STEER	41.2	41.9	45.1
Zeta	GRPO	17.4	22.0	22.3

STEER	19.3	24.0	24.1	
LCB-v5	GRPO	24.4	28.5	29.3
STEER	24.9	29.2	31.8	

Table 5: Performance on real-world coding tasks.

表 5：在现实编码任务中的表现。

sistent with STEER, while the newly introduced hyperparameters are configured following the original implementations, detailed in Appendix E.2.

与 STEER 保持一致，而新引入的超参数则按照原始实现进行配置，详细内容见附录 E.2。

6.2 Main Results

6.2 主要结果

Math Reasoning Tasks. The main results in math reasoning task are shown in Table 3. STEER outperforms classical RLVR baselines as well as existing entropy intervention baselines across all datasets. STEER improves average performance by 2.2 points over the second runner-up (OPO) and by 2.9 points over the third runner-up (Clip-Cov) across all baselines. The performance experiments on Qwen2.5-14B shown in Table 4 are compared with the top three competitors in Table 3 (i.e., OPO, Clip-Cov, and Entro. Adv.). STEER also consistently achieves the highest average performance, demonstrating its superior generalization in improving math reasoning abilities.

数学推理任务。 数学推理任务的主要结果如表 3 所示。STEER 在所有数据集上均优于经典 RLVR 基线以及现有的熵干预基线。STEER 在所有基线中，相对于第二名 (OPO) 提升了 2.2 分，相对于第三名 (Clip-Cov) 提升了 2.9 分。表 4 中显示的 Qwen2.5-14B 的性能实验与表 3 中的前三名竞争者 (即 OPO、Clip-Cov 和 Entro. Adv.) 进行了比较。STEER 也始终实现了最高的平均性能，展示了其在提升数学推理能力方面的优越泛化能力。

Beyond metric avg@32 and avg@1 in Table 3, the results of Pass@256/512/1024 in math reasoning are shown in Figure 6. As observed, STEER delivers the best Pass@256/512/1024 on both AIME24/25 and the highest average. By explicitly regulating entropy, STEER preserves sufficient exploration to discover more informative trajectories, thereby achieving better performance.

除了表 3 中显示的 metric avg@32 和 avg@1 之外，图 6 展示了数学推理中 Pass@256/512/1024 的结果。如观察所示，STEER 在 AIME24/25 上实现了最佳的 Pass@256/512/1024 表现，并且具有最高的平均值。通过显式地调节熵，STEER 保留了足够的探索能力，以发现更具信息量的轨迹，从而实现了更好的性能。

Coding Tasks. Beyond math reasoning, we evaluate our method on real-world code tasks. Table 5 compares STEER with GRPO on three Qwen2.5-coder models. We observe that STEER exceeds GRPO by at least 1% in each model and test set. These results demonstrate the superior performance

编码任务。 除了数学推理之外，我们在现实世界的代码任务上评估我们的方法。表 5 比较了 STEER 与 GRPO 在三个 Qwen2.5-coder 模型上的表现。我们观察到，在每个模型和测试集上，STEER 都至少超过 GRPO 1%。这些结果展示了 STEER 的卓越性能。

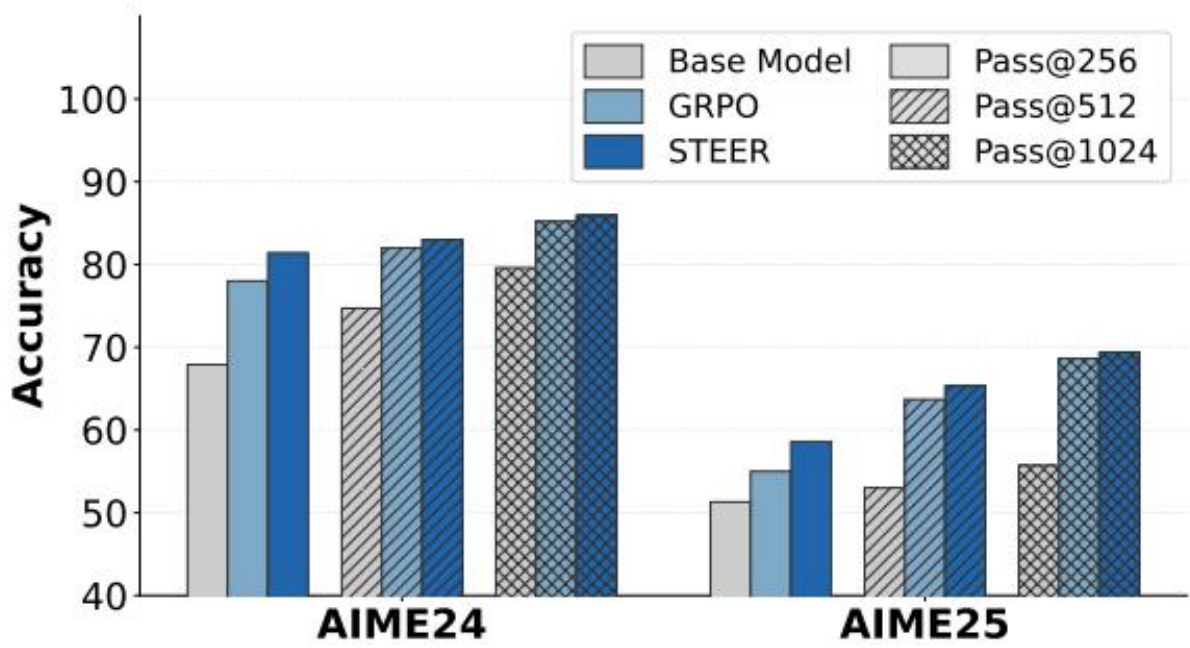


Figure 11: images/ef56d06d78f4de1cefa8a28b19d9c31d0b38fb7e897b4e09f1b4a601a54c0a3b.jpg

Figure 6: Pass@256/512/1024 performance on AIME24 and AIME25.

图 6: AIME24 和 AIME25 上的 Pass@256/512/1024 性能表现。

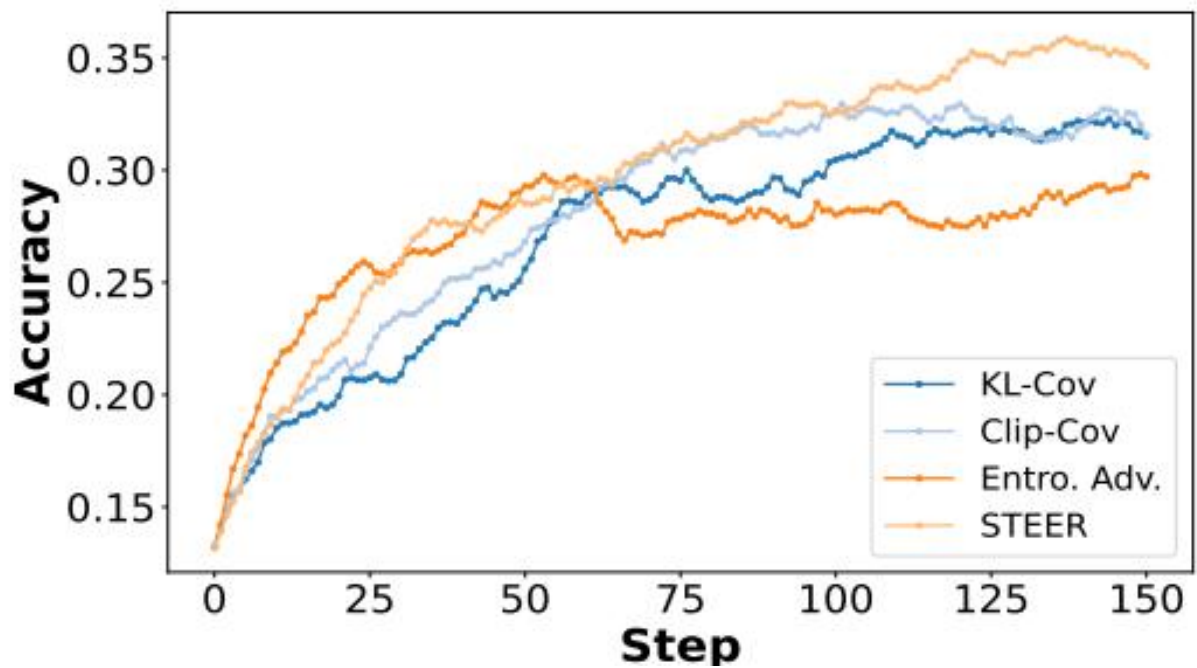


Figure 12: images/76f4fe3650c63aa43111edaebf3b77d2d16662d80c3af29eb3f06aa5e8678e35.jpg

Figure 7: Test accuracy dynamics comparison.

图 7: 测试准确率动态对比。

of STEER in various scenarios. More performance evaluation is detailed in Appendix F.

STEER 在各种场景中的表现。更多性能评估详情请参见附录 F。

Training Dynamics. Figure 7 shows the curves of test accuracy during training. Notably, STEER exhibits a stable upward trajectory, ultimately achieving superior final performance compared to the baselines.

训练动态。图 7 展示了训练过程中测试准确率的变化曲线。值得注意的是，STEER 表现出稳定的上升趋势，最终在性能上优于基线模型。

6.3 Empirical Results for Entropy Modulation

6.3 熵调制的实证结果

To better illustrate STEER’s ability to regulate entropy across a wide range, we adopt an extreme training setup with $\varepsilon_{\text{high}} = 5$ and $\varepsilon_{\text{low}} = 0.99$, where almost no ratio clipping is applied. In such scenarios, RL training is vulnerable due to unstable gradient updates under extreme clipping ratios. The results are shown in Figure 8. Most methods fail to maintain stable entropy: GRPO and Entro. Adv. tend toward entropy collapse; adding an Entropy Loss drives entropy up rapidly, leading to excessive uncertainty; and Clip-Cov cannot reliably control entropy. By contrast, STEER stabilizes after an initial decline and maintains steady entropy subsequently. We test entropy intervention methods in this training setup and test set accuracy shown in

为了更好地说明 STEER 在广泛范围内调节熵的能力，我们采用了一种极端的训练设置，即 $\varepsilon_{\text{high}} = 5$ 和 $\varepsilon_{\text{low}} = 0.99$ ，其中几乎不进行比率裁剪。在这种情况下，由于在极端裁剪比率下梯度更新不稳定，强化学习训练容易受到损害。结果如图 8 所示。大多数方法无法维持稳定的熵：GRPO 和 Entro. Adv. 趋向于熵崩溃；添加熵损失会使熵迅速上升，导致过度的不确定性；而 Clip-Cov 无法可靠地控制熵。相比之下，STEER 在初始下降后趋于稳定，并随后保持稳定的熵。我们在这种训练设置下测试了熵干预方法，测试集准确率如所示。

Method	AIME24	AIME25	AMC23	MATH500	Minerva	Olympiad	Avg.
GRPO	31.6	12.8	66.7	79.0	39.3	40.1	44.9
Entro. Adv.	34.8	13.4	64.3	77.6	37.6	39.9	44.6
Entro. Loss	32.7	14.7	71.3	79.0	36.8	41.4	46.0
Clip-Cov	30.4	14.0	72.3	79.6	37.1	41.7	45.8
STEER (Ours)	36.1	16.0	76.3	80.5	39.5	42.3	48.5

方法	AIME24	AIME25	AMC23	MATH500	弥涅尔瓦	奥林匹克	平均值
GRPO	31.6	12.8	66.7	79.0	39.3	40.1	44.9
进入。副词。	34.8	13.4	64.3	77.6	37.6	39.9	44.6
进入。损失	32.7	14.7	71.3	79.0	36.8	41.4	46.0
剪辑覆盖	30.4	14.0	72.3	79.6	37.1	41.7	45.8
STEER (Ours)	36.1	16.0	76.3	80.5	39.5	42.3	48.5

Table 6: Performance on test datasets in extreme scenarios.

表 6: 在极端场景下的测试数据集性能。

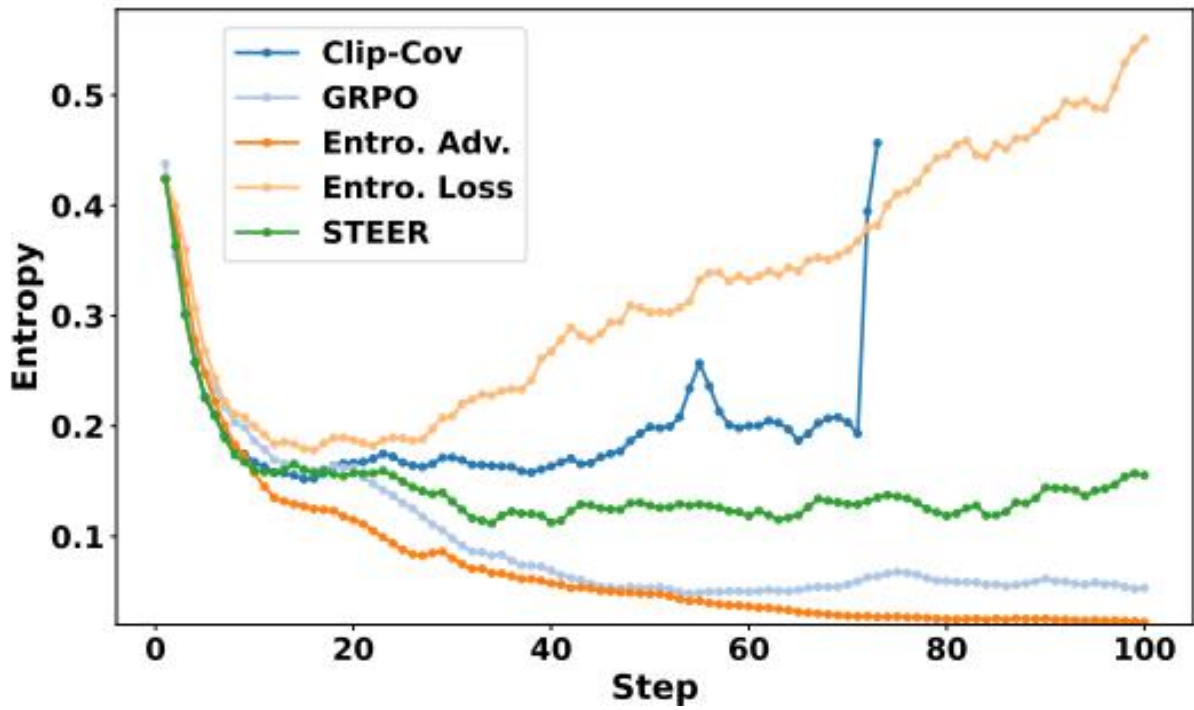


Figure 13: images/ca4520b23362d77328ae43a64434819e0eecbcbf51ad02072e281e95b5235080.jpg

Figure 8: Entropy dynamics in extreme scenarios.

图 8：极端情景下的熵动态变化。

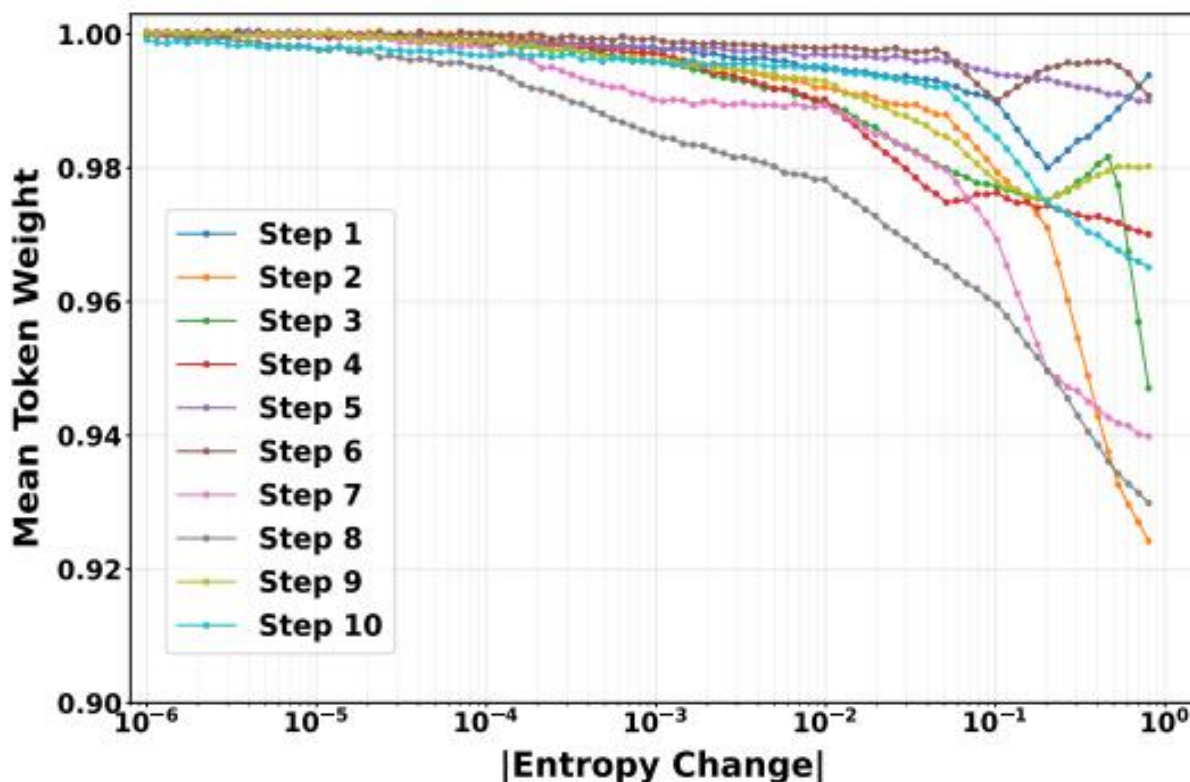


Figure 14: images/f9b06515a07a7792fe40fe04ebe56beee0a505d884b0c94e4a13a9f629dfcc95.jpg

Figure 9: Relationship between mean token weight and entropy change across steps.

图 9：平均标记权重与步骤间熵变的关系。

Table 6. It can be observed that, even in training scenarios where the clipping operation is almost completely removed, STEER maintains relatively stable performance compared to other entropy intervention methods and achieves the highest accuracy across all test sets.

表 6。可以观察到，即使在几乎完全移除裁剪操作的训练场景中，STEER 相比于其他熵干预方法仍保持相对稳定的性能，并在所有测试集上实现了最高的准确率。

Besides, Figure 9 depicts the trend of average token weight as a function of the absolute token entropy change in the first 10 steps. When entropy changes are small, most weights remain near 1; only tokens with large entropy changes receive substantially reduced weights, indicating that STEER stabilizes training without impeding learning.

此外，图 9 展示了平均 token 权重随前 10 步中绝对 token 熵变化的趋势。当熵变化较小时，大多数权重保持在接近 1 的水平；只有那些熵变化较大的 token 会显著降低权重，这表明 STEER 在不阻碍学习的情况下稳定了训练。

6.4 Generalization Study

6.4 普遍性研究

We conduct experiments on additional backbones, including Llama-3.2-3B-Instruct (Grattafiori et al., 2024) for math reasoning and Mistral-7B-v0.3 (Jiang et al., 2023) for code editing. For each task, we use the same STEER hyperparameter as in the main experiments, rather than re-tuned

per backbone. The results are reported in Appendix F.3, and STEER yields consistent performance improvements. We also evaluate STEER’s performance

我们在其他主干网络上进行实验，包括用于数学推理的 Llama-3.2-3B-Instruct (Grattafiori 等, 2024) 和用于代码编辑的 Mistral-7B-v0.3 (Jiang 等, 2023)。对于每个任务，我们使用与主实验中相同的 STEER 超参数，而不是为每个主干网络重新调优。结果在附录 F.3 中报告，STEER 带来了一致的性能提升。我们还评估了 STEER 的性能

with varying model size including 1.5B/7B/14B, as reported in Table 3, Table 4 and Appendix F.3. We observe STEER yields superior performance across multiple model sizes.

具有不同模型规模，包括 1.5B/7B/14B，如表 3、表 4 和附录 F.3 所示。我们观察到 STEER 在多个模型规模上均表现出卓越的性能。

Besides, we evaluate STEER under two other representative RL algorithms, RLOO (Ahmadian et al., 2024) and OPO (Hao et al., 2025) with the default setting in verl implementations. The results are also shown in Appendix F.3. STEER improves the average score from 45.8 $\rightarrow$ 46.8 on RLOO and 46.4 $\rightarrow$ 47.5 on OPO, indicating that STEER can be effectively applied to other RLVR algorithms rather than specific to GRPO.

此外，我们在两个其他具有代表性的强化学习算法 RLOO (Ahmadian 等, 2024) 和 OPO (Hao 等, 2025) 上评估 STEER，使用 verl 实现中的默认设置。结果也显示在附录 F.3 中。STEER 将 RLOO 上的平均得分从 45.8 $\rightarrow$ 46.8 提高到了 46.4 $\rightarrow$ 47.5，表明 STEER 可以有效地应用于其他 RLVR 算法，而不仅限于 GRPO。

7 Conclusion

7 结论

In this work, we conduct comprehensive theoretical and empirical analyses of entropy dynamics in RLVR. We derive a tight approximation for token-level entropy change at each update step, revealing four governing factors and providing a unified view to demystify the mechanisms of existing strategies. We further introduce STEER, a principled entropy-modulation strategy that adaptively reweights tokens based on the theoretically-estimated entropy variations.

在本工作中，我们对 RLVR 中的熵动态进行了全面的理论和实证分析。我们推导出一个紧密的近似值，用于描述每个更新步骤中令牌级别的熵变化，揭示了四个主导因素，并提供了一个统一的观点，以解开现有策略的机制。我们进一步引入了 STEER，这是一种基于理论估计的熵变化来自适应地重新加权令牌的熵调节策略。

8 Limitations

8 局限性

This study focuses on settings with verifiable rewards, while RL training without verifiable rewards is not explored. Extending our analysis to more general scenarios, such as environments with limited verifiability, would be an interesting direction for future work. It would be also valuable to investigate whether our findings generalize to other reinforcement learning settings, such as agentic and long-horizon RL.

本研究关注的是具有可验证奖励的环境，而没有可验证奖励的强化学习训练并未被探讨。将我们的分析扩展到更一般的场景，例如可验证性有限的环境，将是未来工作的有趣方向。此外，研究我们的发现是否可以推广到其他强化学习设置，例如自主代理和长时间地平线的强化学习，也将具有重要价值。

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